LINUX KERNEL /DEV/UHID | FIDO2 / CTAP2 STANDARD | ANDROID STRONGBOX

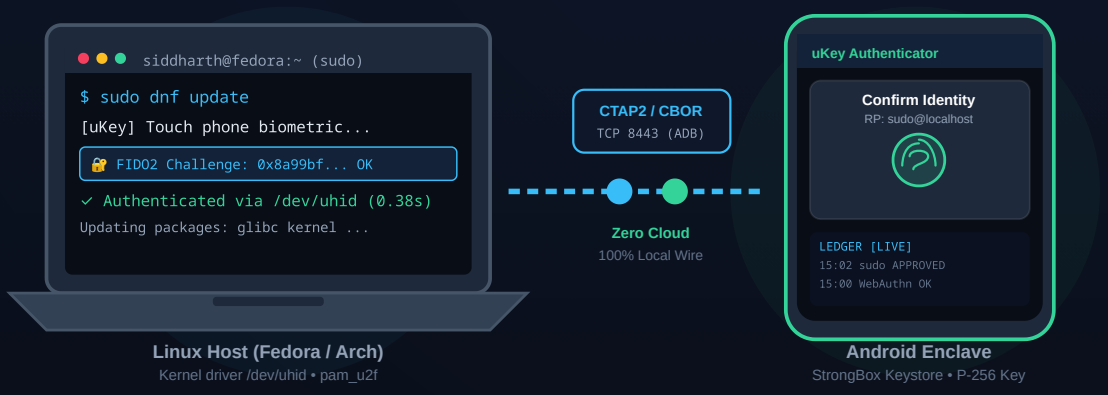
uKey: Virtual Hardware Security Key

Turns the smartphone already in your pocket into a bank-grade FIDO2/CTAP2 USB security key for Linux. Eliminates passwords for `sudo`, OpenSSH, and WebAuthn with 0.0s instant bypass and zero cloud dependencies.

$0	<400ms	0.0s	100%
HARDWARE COST (VS $75 YUBIKEY)	ECDSA P-256 AUTH SPEED	INSTANT ENTER SKIP DELAY	LOCAL / ZERO CLOUD TRACKING

HARDWARE TOPOLOGY & CRYPTOGRAPHIC CHANNEL

• POINT-TO-POINT LOCAL BRIDGE ACTIVE



Linux Host (Fedora / Arch)
Kernel driver /dev/uhid • pam_u2f

Android Enclave
StrongBox Keystore • P-256 Key

🧒 The 30-Second "Clubhouse" Analogy (Explain Like I'm 5)

Imagine your Linux laptop is a high-security treehouse. Normally, you have to type a 20-letter secret spell 40 times a day to enter (password). Other people pay $75 for a tiny plastic key (YubiKey) that they often forget or lose. **uKey uses the magic phone already in your pocket.** Whenever you want to open the door, your laptop whispers to your phone. You tap your thumb, and the door opens instantly in 0.4 seconds!

How It Works: Under The Hood

uKey bypasses complex, brittle user-space hacks by speaking directly to the Linux Kernel via user-space HID emulation (`/dev/uhid`), making the phone appear as a certified USB security dongle.

LAYER 01 • LINUX KERNEL

🐧 User-Space HID Emulation

The Linux daemon opens `/dev/uhid` and registers a USB HID device with the standard FIDO2 usage page (`0xF1D0`). To Google Chrome, Firefox, OpenSSH, and PAM, your system sees a genuine physical USB token plugged into a motherboard port.

```
HID Report Descriptor: 0xF1D0 (FIDO Alliance)
Packet Framing: 64-byte CTAPHID (Init & Cont)
```

LAYER 02 • ANDROID HARDWARE

🔒 StrongBox KeyStore Silicon

Cryptographic keypairs are generated inside the phone's dedicated hardware security module (StrongBox or ARM TrustZone TEE). The private key **never enters device RAM** and cannot be extracted even by a root user or kernel debugger.

```
Key Type: ECDSA NIST P-256 (COSE -7)
Hardware Guard: KeyProperties.PURPOSE_SIGN
```

THE 3-STEP CRYPTOGRAPHIC HANDSHAKE

1. **Challenge Generation (Linux Host)**
   When you trigger `sudo` or a WebAuthn portal, `pam_u2f` or the browser generates a 32-byte cryptographic challenge (SHA-256 clientDataHash) and sends it across the virtual USB pipe via CTAP2 `authenticatorGetAssertion` .

2. **Out-of-Band Mobile Prompt (Android Phone)**
   The phone receives the challenge over the local socket. The Android OS displays native `BiometricPrompt` showing the exact requesting application. Biometrics are verified in hardware; no fingerprint data leaves silicon.

3. **Hardware Signature & Instant Unlock (<400ms)**
   The StrongBox chip calculates an ECDSA signature over `authenticatorData || clientDataHash` with User Present (UP=1) and User Verified (UV=1) flags. The assertion returns through `/dev/uhid` , and Linux grants access.

Privacy & Zero-Trust Guarantees

- **Zero Biometrics Transferred:** Fingerprint minutiae templates remain permanently locked inside the phone's chip. Only ephemeral digital signatures cross the wire.
- **Zero Third-Party Relays:** Works 100% offline via local USB cable or loopback Wi-Fi. No Google, AWS, or cloud relays.
- **Unphishable:** FIDO2 mathematically binds credentials to the domain origin. Malicious phishing links cannot clone assertions.

Signature Capabilities

uKey was engineered to solve real-world usability friction that traditional security keys ignore.

⚡ 0.0s Instant Password Skip

Don't have your phone beside you? **Press Enter** or `Ctrl+C`. In 0 milliseconds, PAM falls back to standard UNIX password entry. Zero timeouts, zero waiting.

📴 Proximity Walk-Away Lock

Unplug your phone to grab coffee, and uKey detects bridge disconnect in 50ms, triggering `loginctl lock-session`. Never leave your desktop exposed.

📜 Out-of-Band Audit Ledger

Every single `sudo` or WebAuthn attempt is echoed to your phone screen in real time with timestamp and domain. Functions as an instant intrusion detector.

🖐️ Adaptive Sensor Detection

Laptop has a palm-rest fingerprint reader? uKey detects it via `libfprint` and co-exists cleanly. Laptop has no sensor? uKey makes your phone primary.

Market Comparison Matrix

Security Vector / Feature	Traditional Passwords	Physical YubiKey ($75)	uKey (Our Project)
Hardware Cost	$0	$70 – $90 / key	**$0 (Uses existing phone)**
Kernel-Level USB Device Emulation	✗	✓ (Fixed silicon)	✓ **Native** `/dev/uhid`
Linux Terminal `sudo` & SSH	✓ (Exposed text)	✓ (Blind touch)	✓ **Biometric + Context**
Phishing Defense (FIDO2)	✗ (High risk)	✓	✓ **Cryptographically Bound**
Screen Auto-Lock on Walk-Away	✗	✗ (Requires custom rules)	✓ **Built-in** `loginctl`
Real-Time Mobile Audit Log	✗	✗ **(No screen on key)**	✓ **Live Audit Trail**

THE 60-SECOND INVESTOR & CONFERENCE PITCH

> *"Over 90% of Linux developers use laptops without biometric sensors or hardware security enclaves. Buying $75 hardware YubiKeys is expensive and inconvenient for teams.*
>
> *We created **uKey**: an open-source bridge that transforms the smartphone already in your pocket into a virtual USB FIDO2/CTAP2 security key. Operating directly at the Linux kernel level via `/dev/uhid`, uKey enables instant biometric authentication for terminal `sudo`, OpenSSH, and browser WebAuthn logins in 400ms using Android's hardware StrongBox enclave. With zero cloud dependencies, proximity walk-away locking, and zero-wait password skip, uKey makes zero-trust desktop security universal and free."*